Oniris
VetAgroBio Nantes
ÉCOLE NATIONALE

Journée de la recherche

LLMs applications in a scientific research context

Presented by Joseph Allyndrée – 11/07/2024

01
About prompting

‘How I learned to start worrying about prompt formatting’

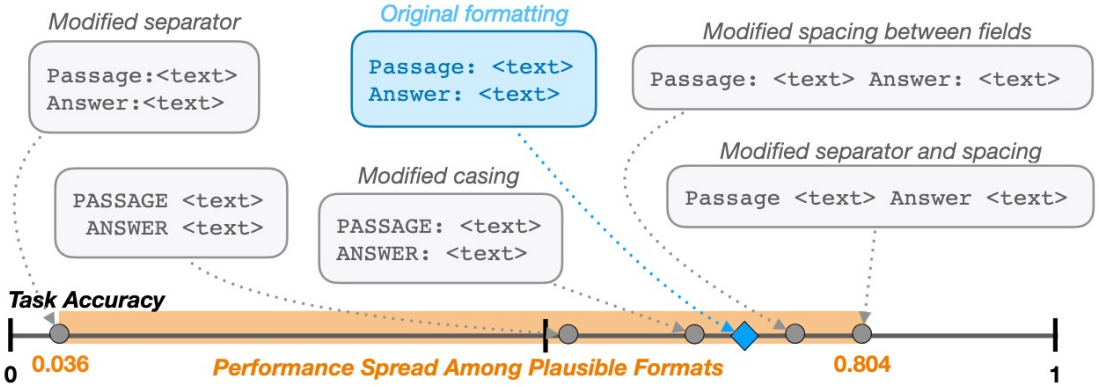


Figure 1: Slight modifications in prompt format templating may lead to significantly different model performance for a given task. Each <text> represents a different variable-length placeholder to be replaced with actual data samples. Example shown corresponds to 1-shot LLaMA-2-7B performances for task280 from SuperNaturalInstructions (Wang et al., 2022). This StereoSet-inspired task (Nadeem et al., 2021) requires the model to, given a short passage, classify it into one of four types of stereotype or anti-stereotype (gender, profession, race, and religion).

Sclar, Melanie, et al. "Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting." arXiv preprint arXiv:2310.11324 (2023).

How to prompt ?

« Prompt engineering » is a real thing !
With prompting, you give guardrails to LLMs and possibly better reasoning capacities

- Be clear and direct
- Give examples of what you expect
- Assign a role to the LLM (you are …, you are very good at …)
- Use chain of thought (do this, then that …)
- Last but not least : think outside of the box !

https://platform.openai.com/docs/guides/prompt-engineering
https://cloud.google.com/gemini/docs/discover/write-prompts
https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview
Enhance your prompts : https://neuralwriter.com/prompt-tool/

Enhance it with me

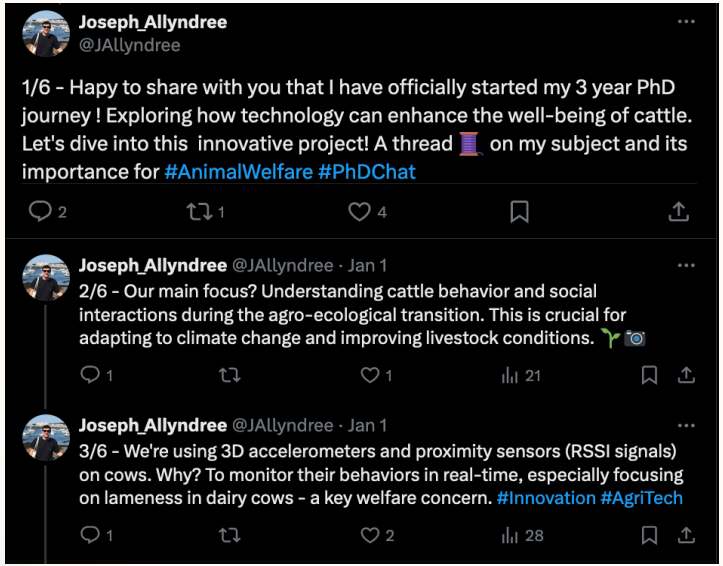
Be clear and direct
Give examples
Assign a role
Use chain of thought

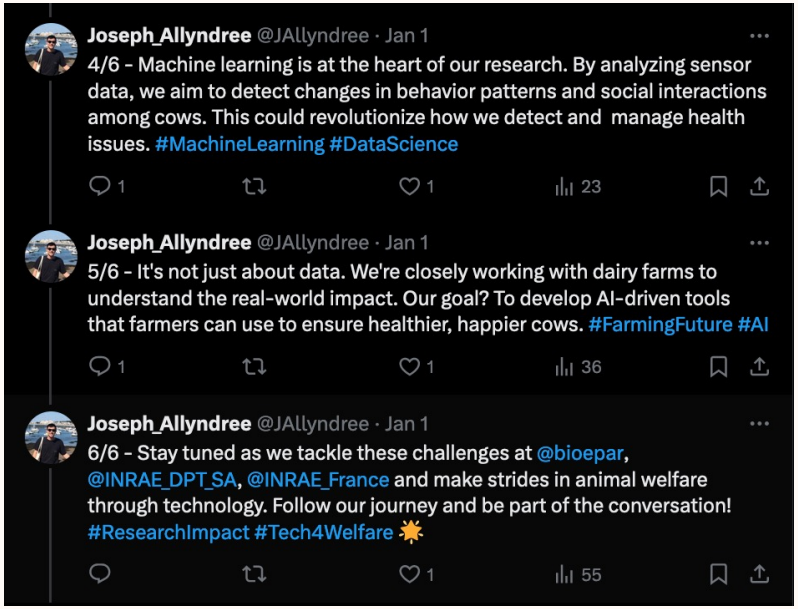
« Write an engaging tweet »

⬇

”You are now a twitter expert, managing a research account with high engagement. You are a communication god and use twitter to talk about your research
Write a 280-character tweet that effectively conveys a clear and compelling message in a conversational tone, using relevant hashtags to maximize reach and impact Please include a creative element that makes it engaging when you see fit
Once you have created the tweet, read it again and modify spelling and grammar mistakes”

A concrete example





02
Local LLMs

Why local ?

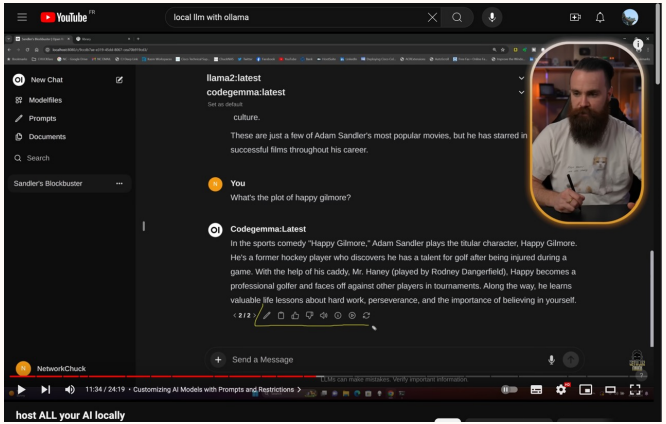
- Privacy and security
- More specialized models !
- Possibility of fine tuning a model with YOUR data
- Data gouvernance
- Available offline
- **Reduce environment impact !!**

➔ **Perfect for research !**

No data leaks to openAI or other providers !
Compromise with less reasoning capacities (not true depending on hardware)
Hardware dependent !



How to local LLMs ?



https://www.youtube.com/watch?v=Wjrdr0NU4Sk

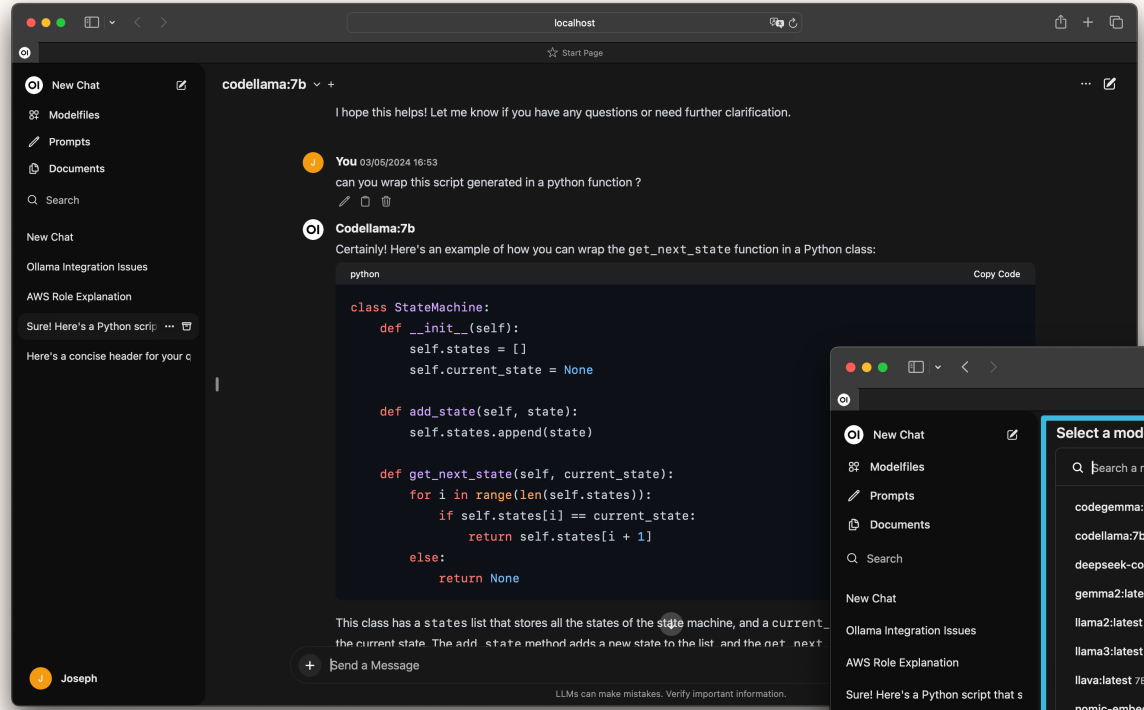


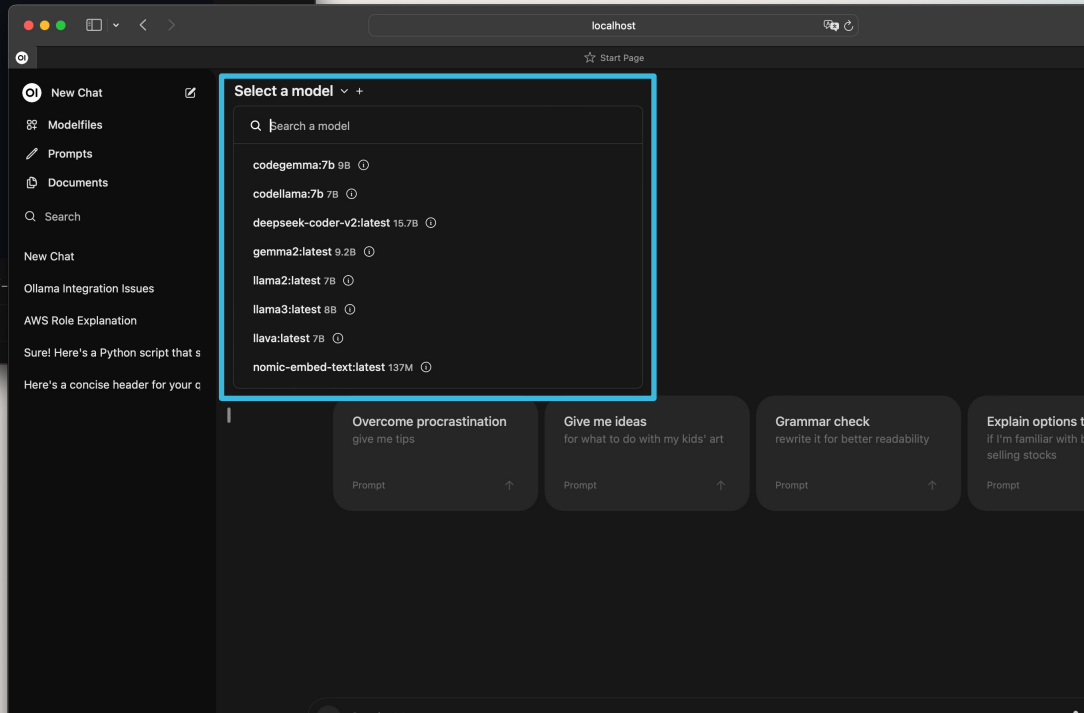
https://openwebui.com/

https://ollama.com/

https://www.doprax.com/tutorial/a-step-by-step-guide-for-installing-and-running-ollama-and-openwebui-locally-part-1/

https://docs.openwebui.com/getting-started/


localhost
Start Page
New Chat
Modelfiles
Prompts
Documents
Search
New Chat
Ollama Integration Issues
AWS Role Explanation
Sure! Here's a Python scrip
Here's a concise header for your c
Joseph
codellama:7b
I hope this helps! Let me know if you have any questions or need further clarification.
You 03/05/2024 16:53
can you wrap this script generated in a python function ?
Codellama:7b
Certainly! Here's an example of how you can wrap the get_next_state function in a Python class:
python
Copy Code
class StateMachine:
def __init__(self):
self.states = []
self.current_state = None
def add_state(self, state):
self.states.append(state)
def get_next_state(self, current_state):
for i in range(len(self.states)):
if self.states[i] == current_state:
return self.states[i + 1]
else:
return None
This class has a states list that stores all the states of the state machine, and a current_
Send a Message
LLMs can make mistakes. Verify important information.


localhost
Start Page
New Chat
Modelfiles
Prompts
Documents
Search
New Chat
Ollama Integration Issues
AWS Role Explanation
Sure! Here's a Python script that s
Here's a concise header for your c
Select a model
Search a model
codegemma:7b 9B
codellama:7b 7B
deepseek-coder-v2:latest 15.7B
gemma2:latest 9.2B
llama2:latest 7B
llama3:latest 8B
llava:latest 7B
nomic-embed-text:latest 137M
Overcome procrastination
give me tips
Prompt
Give me ideas
for what to do with my kids' art
Prompt
Grammar check
rewrite it for better readability
Prompt
Explain options t
if I'm familiar with b
selling stocks
Prompt

Thanks!

Do you have any questions?

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@Jallyndree (Twitter)

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